

Weiwu Zhu

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Employment

Apple Engineering Manager
Seattle, WA, United States Nov. 2022 - Present

- Leading a machine learning engineering team focusing on **AI Agent** based question answering system for **Siri** as a manager. My team optimized a bunch of critical components, such as embedding models for **RAG** and query understanding, **LLM** based planner and summarizer, and synthetic data generator via distillation.

Meta Machine Learning Engineer
Bellevue, WA, United States Jan. 2020 - Nov. 2022

- Worked on **Ads Ranking** and Delivery for Facebook App Monetization team as a **Tech Lead**. Leading the whole engineering team to optimize monetization efficiency on FB video surfaces like Reels and Story through modeling, signal mining, delivery optimization, etc.

Microsoft Senior Software Engineer
Bellevue, WA, United States Jan. 2015 - Dec. 2019

- Worked on Bing **Local Entity Search** as a **Tech Lead**, especially on entity understanding by leveraging **Deep Learning**, query understanding with **Natural Language Processing (NLP)**, search relevance, and large scale data mining and crawling.

Microsoft China Software Engineer
Beijing, China Jul. 2011 - Dec. 2014

- Worked on Microsoft virtual assistant **Cortana** in many different areas, including **Speech Synthesis (TTS)** engine, **Natural Language Generation (NLG)**, and **Knowledge Graph** cataloging billions of entities and their relationships.

Experience

Microsoft Research Asia Intern Researcher
Beijing, China Jan. 2010 - Jun. 2011

- Research project: Voice Activity Detection for speech recognition in noisy environments.

Patent

Weiwu Zhu, jointly with Albert J. Thambiratnam & Frank T. Seide, Leveraging speech recognition feedback for voice activity detection, *United States Patent No. 8,650,029*

Weiwu Zhu, jointly with Ming Sui & Shengcai Peng, Language Modeling For Speech Recognition Leveraging Knowledge Graph, *WO2016196320*

Publication

Feijie Wu, **Weiwu Zhu**, et al. PORTool: Importance Aware Policy Optimization with Rewarded Tree for Multi Tool Integrated Reasoning, *arXiv:2510.26020, NeurIPS 2026 submitted*

Weiwu Zhu, Using Knowledge Graph and Search Query Click Logs in Statistical Language Model For Speech Recognition, *InterSpeech 2017*

Kit Thambiratnam, **Weiwu Zhu**, and Frank Seide, Voice Activity Detection Using Speech Recognizer Feedback, *InterSpeech 2012*

Skill

Programming: Skilled in Python and C++, and familiar with C#, Java, SQL, and Bash; Professional at Search Relevance, Ads Ranking, Speech Technology, and Machine Learning;
Language: Chinese (native), English (fluent)

Education

Peking University, Beijing, China Sep. 2008 - Jun. 2011
M.S. in Software Engineering

Wuhan University, Wuhan, China Sep. 2004 - Jun. 2008
B.S. in Software Engineering